JACOB FERNANDEZ

Gameplay Programmer | Software Engineer

San Antonio, TX | Email | GitHub | LinkedIn | Portfolio

EDUCATION

University of the Incarnate Word, San Antonio, TX Expected 2026

Accelerated B.F.A./M.F.A., Game Programming: M.F.A. (2024–2026), B.F.A. with Minor in Computer Interactive Systems (2021–2026)

- Honors: Triple A Programmer Award, 2024 | Relevant Coursework: Advanced C++, Game Engine Architecture, Computer Graphics, AI Systems, Data Structures & Algorithms, Database Systems

TECHNICAL SKILLS

Languages: C++, C#, Python, JavaScript, TypeScript, HTML/CSS

Engines & Frameworks: Unreal Engine 5, Unity 6, OpenGL, Blueprint, Gameplay Ability System, Enhanced Input, Niagara VFX, Cinemachine, NavMesh, Crow (C++ REST), Next.js, PyQt6

Tools/Developer Environment: Git, Perforce, CMake, Visual Studio, VS Code, Rider, RenderDoc

EXPERIENCE

Game Programming Teaching Assistant, UIW Sep 2025 – Present

- Taught advanced **C++** and **Unreal Engine 5** to students transitioning from C#/Unity, covering systems programming, memory management, and design patterns
- Developed curriculum (lecture decks, coding labs, project specs) spanning pointers, file I/O, and gameplay systems; mentored students on debugging, code review, AI behavior trees, and Gameplay Ability System architecture

Graduate Game Developer, MFA Prototyping Program, UIW 2024 – Present

- Collaborated on a cross-functional team to ship multiple game prototypes from concept through playable build; currently engineering a turn-based horror game in **Unreal Engine 5.6** for Summer 2026, working on core gameplay systems and technical design

PROJECTS

Hack & Slash Combat System | *UE5.7, C++, Blueprint, Niagara* GitHub

- Engineered a Devil May Cry/FFXVI–inspired combo system with buffered-input cancels, 4-hit light/heavy chains, and progressive-gravity air combos; designed a style-rank meter, dynamic lock-on camera, homing projectile, and enemy-pull ability

OpenGL Rendering Engine | *C++17, OpenGL 4.1, GLSL, Lua* GitHub

- Built a real-time rendering engine from scratch supporting 8 simultaneous dynamic lights and real-time cubemap reflections with Fresnel-based chrome shading; implemented Lua-driven scene scripting and a full ImGui editor with Maya-style viewport navigation

Metal Gear Mechanics Clone | *Unity, C#, NavMesh, Cinemachine* GitHub

- Recreated core MGS1 stealth systems: 4-state AI behavior machine (Patrol/Caution/Aggressive/Search), CQC grab mechanics, a cardboard-box disguise system, and a dynamic wall-cover camera

CS2 Skin Market API | *C++17, Crow, libcurl, JavaScript* GitHub

- Built a REST API serving live Steam Community Market data with a 0/1 knapsack dynamic-programming budget optimizer and a round-robin algorithm for loadout variety; deployed with a live demo

LEADERSHIP / EXTRACURRICULAR

UPGRADE Program Representative, UIW 2022 – 2025

- Presented C++, C#, Unreal Engine, and Unity workflows to prospective students at recruitment events, translating technical concepts for non-technical audiences